DynSGS without and with the smoothness cutoff $c = 10^{-3}$

In the notation of the paper (*The shock capturing model*): equation numbers (11)–(16) and Remark 1 are the paper's, (C), (L), (R) this note's. Measured on the smooth Alfvén wave (`problems/MHD/alfvenWaveBabbarRanocha2026`; Babbar & Ranocha 2026, §5.5.1), branch `sm/MHD`, commit `e40bc3e`. The cutoff is the deck entry `AW_CUTOFF` (`:dsgs_cutoff`), 10^{-3} by default.

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1. The coefficient, and where the cutoff enters. With $\Delta_e = h_e/(N+1)$ (15), $C_1 = 1$, $C_2 = \frac{1}{2}$, and $\mu_{\max}|_e$, $\mu|_e$ as in (13)–(14), the code computes

$$\mu_{\text{res}}\big|_e = C_1\,\Delta_e^2\,\max\big(0,\ r_e - c\big), \qquad r_e = \max_{i\in\mathcal{I}} \frac{\|R_i\|_{L^\infty(\Omega_e)}}{\max\big(\|q_i - \langle q_i\rangle\|_{L^\infty(\Omega)},\ \sigma_i\big)}. \tag{C}$$

The two maxima run over different indices. As in (11)–(12), i is the *equation* and p the *node*: the norm $\|R_i\|_{L^\infty(\Omega_e)} = \max_{p\in\Omega_e}|R_{i,p}|$ is the largest residual of equation i on the element, and $\max_{i\in\mathcal{I}}$ is the largest over the equations ($\mathcal{I}$: the eight conservation laws, ψ excluded). Each equation is divided by its own scale first, which makes all eight rates (1/time) and so comparable; $\mu|_e$ is one coefficient for all of them. **Without the cutoff**, $c = 0$, (C) is (12) with the floors (16). **With it**, $c = 10^{-3}$ is subtracted from the normalized residual r_e first, so $\mu|_e = 0$ wherever $r_e \le c$, and $\mu|_e$ is continuous across it (not an on/off switch). The floors are those of (16) with $\epsilon_{\text{rel}} = 1$ (`:dsgs_rel`; (16) says 10^{-3}): $\sigma_\rho = \bar\rho$, $\sigma_{\rho\boldsymbol{u}} = \bar\rho\bar c$, $\sigma_E = \bar\rho\bar c^{\,2}$, $\sigma_{\boldsymbol{B}} = \sqrt{\bar\rho}\,\bar c$. For this wave $\bar\rho = 1$ and $\bar c = \sqrt{\gamma\bar p/\bar\rho} = 0.42$, the sound speed of the mean state ($\bar p = 0.107$, from $\langle E\rangle$, $\langle\rho\boldsymbol{u}\rangle$, $\langle\boldsymbol{B}\rangle$). The wave's spread ($\le 0.1$) is below the floors, so the momentum and $\boldsymbol{B}$ residuals are divided by $\sigma = \bar\rho\bar c = 0.42$. Since $\rho \equiv 1$, the max over i is decided by the transverse components, ρw and B_z (amplitude $A = 0.1$).

2. The two sensors: what R_i is. "Legacy" is the code's name for the sensor of the runs before September 2026; "residual" is the default since. Both evaluate a time difference minus a spatial rate, on the step-cadenced history of Remark 1 ($\boldsymbol{q}^n, \boldsymbol{q}^{n-1}, \boldsymbol{q}^{n-2}$, rolled once per step). Let $\boldsymbol{q}^{(s)}$ be the Runge–Kutta stage state at $t^n + \theta_s\Delta t$ ($\theta_1 = 0$, so $\boldsymbol{q}^{(1)} = \boldsymbol{q}^n$), $\mathcal{R}_{e,i}$ the bracket of the semi-discrete statement $M\,d\boldsymbol{q}_h/dt = \mathcal{R}(\boldsymbol{q}_h)$ *before* $\bigoplus_e$ (the element's own right-hand side), and M_e the element's lumped mass. The *legacy* sensor is (11) at every stage, the stage state taking the place of $\boldsymbol{q}^n$; the *residual* sensor is (11) at the first stage only, with the element's own rate in place of the assembled one:

$$\text{legacy:}\quad R_i\big|_p = \frac{3q_{i,p}^{(s)} - 4q_{i,p}^{n} + q_{i,p}^{n-1}}{2\Delta t} - \big(M^{-1}\mathcal{R}_i(\boldsymbol{q}^{(s)})\big)_p \qquad \text{at every stage } s, \tag{L}$$

$$\text{residual:}\quad R_i\big|_p = \frac{3q_{i,p}^{n} - 4q_{i,p}^{n-1} + q_{i,p}^{n-2}}{2\Delta t} - \big(M_e^{-1}\mathcal{R}_{e,i}(\boldsymbol{q}^{n})\big)_p \qquad \text{at } s = 1;\ \mu|_e \text{ held for the step.} \tag{R}$$

	legacy (L), `:dsgs_sensor => "legacy"`	**residual (R)**, `"residual"`, the default
time difference	BDF2 weights on $(\boldsymbol{q}^{(s)}, \boldsymbol{q}^n, \boldsymbol{q}^{n-1})$: $\frac{1}{2}(3\theta_s - 1)\,\partial_t\boldsymbol{q}$	BDF2 of three step states: $\partial_t\boldsymbol{q} - \frac{\Delta t^2}{3}\,\partial_t^3\boldsymbol{q}$
spatial rate	$M^{-1}\mathcal{R}$, assembled: the integrator's own $\partial_t\boldsymbol{q}_h$	$M_e^{-1}\mathcal{R}_e$, the element's own: $-(\nabla\cdot\boldsymbol{F}_h - \boldsymbol{s})\vert_{\Omega_e}$
$\mu\vert_e$ computed	at every stage	once per step (`:dsgs_freeze_stage`, on by default)
R_i on a smooth flow	$\frac{3}{2}(\theta_s - 1)\,\partial_t q_i$: how fast q_i changes	$-\frac{\Delta t^2}{3}\partial_t^3 q_i$, plus face jumps $O(h^N)$
$\mu_{\text{res}}\big\vert_e$ on a smooth flow	$O(h^2)$, whatever N	$O(h^4) + O(h^{N+2})$ at a fixed Courant number
MHD decks	Orszag–Tang, rotor, blast, both flux-emergence	Alfvén wave, smooth vortex, Brio–Wu, jet

What (L) measures. Since $M\,d\boldsymbol{q}_h/dt = \mathcal{R}$, $\big(M^{-1}\mathcal{R}_i(\boldsymbol{q}^{(s)})\big)_p$ *is* the rate $\partial_t q_{i,p}$ the integrator uses at that stage, so (L) has no spatial term left: it is the error of its time difference as an estimate of $\partial_t q$, a property of the solution's time history and not of its spatial resolution. And that difference is not $\partial_t q$: its newest point $\boldsymbol{q}^{(s)}$ is $\theta_s\Delta t$ after $\boldsymbol{q}^n$, not Δt. Expanding about t^n, $(3q^{(s)} - 4q^n + q^{n-1})/(2\Delta t) = \frac{1}{2}(3\theta_s - 1)\,\partial_t q + O(\Delta t)$: $-\frac{1}{2}\partial_t q$ at $s = 1$, and $\partial_t q$ only at $\theta_s = 1$. Hence $R_i \approx \frac{3}{2}(\theta_s - 1)\,\partial_t q_i$, the rate at which the solution changes ($\lambda\,\partial_x q$ for a wave), on resolved and unresolved flow alike: a gradient sensor rather than a residual. On this wave $r_e \approx \frac{3}{2}\,\omega A/\sigma \approx 2$ at $s = 1$ ($\omega = 2\pi$, $A = 0.1$), so $\mu_{\text{res}}\big|_e \approx 2\Delta_e^2$, $O(h^2)$ whatever N. $N = 4$ on 16^2 ends with a state error of $3.5\cdot10^{-3}$, against $1.9\cdot10^{-6}$ for (R) without the cutoff, $4.2\cdot10^{-7}$ with it, and $4.3\cdot10^{-7}$ for Galerkin; no cutoff $c \ll 2$ changes that. With a consistent time difference the assembled rate would leave only that difference's truncation error, as small at a shock as on a smooth wave (`sod1d`: μ at the shock 0.5% of the cap): (L) sees a shock only through its stencil's inconsistency.

What (R) measures. With the LGL rule (summation by parts), $\big(M_e^{-1}\mathcal{R}_{e,i}\big)_p = -\big(\nabla\cdot\boldsymbol{F}_{h,i} - s_i\big)\big|_{\Omega_e,p}$, the strong form on Ω_e's own interpolant. At the interior nodes of Ω_e it equals $(M^{-1}\mathcal{R}_i)_p$. At a node shared with $\Omega_{e'}$ it differs from it by the mass-weighted jump $[\![\nabla\cdot\boldsymbol{F}_{h,i}]\!] = \nabla\cdot\boldsymbol{F}_{h,i}|_{\Omega_e} - \nabla\cdot\boldsymbol{F}_{h,i}|_{\Omega_{e'}}$, which the assembly averages away. On a smooth solution, therefore,

$$R_i\big|_p = \underbrace{-\tfrac{\Delta t^2}{3}\,\partial_t^3 q_{i,p}}_{\text{BDF2 truncation error, every node}} + \underbrace{\tfrac{m_{e'}}{m_e + m_{e'}}\,[\![\nabla\cdot\boldsymbol{F}_{h,i}]\!]_p}_{\text{face nodes only}}.$$

The jump is the spatial information the paper describes: $O(h^N)$ where the solution is smooth and resolved, $O(1/h)$ across an under-resolved discontinuity. The first term does not depend on the mesh at a given Δt: it is the floor of §3.

3. Without the cutoff: a floor that converges at order 4. Computing $\mu|_e$ at $s = 1$ only keeps the floor $O(\Delta t^2)$: at a later stage the $O(\Delta t)$ error of the internal stage would enter (order 3.2 instead of 4.8 at $N = 4$). A travelling wave has $\partial_t^3 \boldsymbol{q} = \omega^3 A \neq 0$ however well it is resolved, so the first term of (R) keeps r_e above

$$r_{\text{floor}} = \frac{\Delta t^2}{3} \frac{\omega^3 A}{\bar{\rho}\bar{c}} \quad (4 \cdot 10^{-6} \text{ to } 7 \cdot 10^{-4} \text{ over the sweep}),$$

and $\mu_{\text{res}}\big|_e \approx C_1 \Delta_e^2 r_{\text{floor}} \propto \Delta_e^2 \Delta t^2 \propto h^4$ at a fixed Courant number, *whatever* N. The paper's asymptotic estimate keeps this $O(\Delta t^2)$ in $\|R_i\|_{L^\infty(\Omega_e)}$ but drops it from μ_{res}: $\mu_{\text{res}} = O(h^{N+2}) + O(h^4)$, and the second term wins for $N > 2$. That μ damps the wave by $\mu k^2 t$ ($k = 2\pi$), a state error of $0.14\,\mu k^2 t$:

- it predicts the measured error with no fitted constant ($N = 6$ on 16^2: $5.4 \cdot 10^{-8}$ both; $N = 5, 6, 7$ within 0.95–1.4× from 8^2 on);
- DynSGS converges at ≈ 4 for $N = 5, 6, 7$, and at equal #DOFs the three curves coincide, since Δ_e and Δt are both set by the node spacing;
- halving Δt alone divides the error by 3.9–4.0, and the Galerkin error does not move;
- at $N = 3$ the floor is ∼1/100 of Galerkin's own error, and does not show.

4. With the cutoff $c = 10^{-3}$. $r_{\text{floor}} < 10^{-3}$ on every mesh of the sweep, so $\mu|_e \equiv 0$ and DynSGS *is* the Galerkin scheme: bit for bit, all nine variables, L^2 and L^∞, for $N = 5, 6, 7$ on every mesh but the coarsest of $N = 5$. The orders are Galerkin's (6.5, 6.0, 8.6 on the finest pair). Where the residual is real the cutoff barely matters: at a discontinuity $r_e = O(10^2)$, which c changes by 10^{-5} relative.

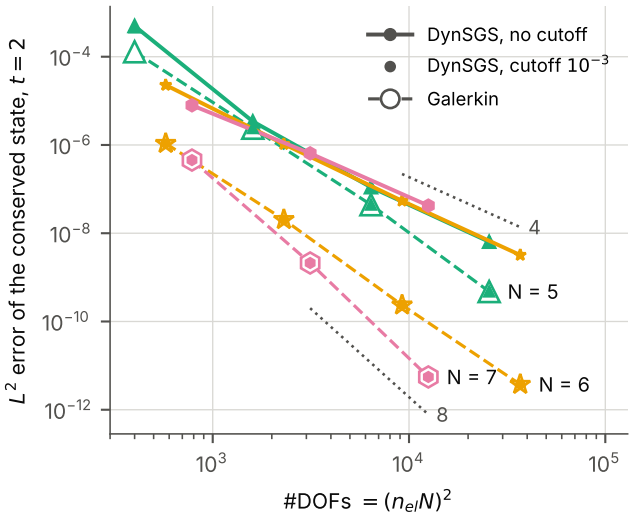


Figure 1: State error at $t = 2$, $N = 5, 6, 7$. Solid: DynSGS (R) without the cutoff (slope 4). Small filled markers inside the hollow Galerkin ones: DynSGS (R) with $c = 10^{-3}$, identical to Galerkin.

N	mesh	no cutoff		$c = 10^{-3}$		Galerkin	
5	8^2	3.4e-6		2.4e-6		2.4e-6	
5	16^2	1.1e-7	5.0	4.4e-8	5.8	4.4e-8	5.8
5	32^2	6.4e-9	4.1	4.8e-10	6.5	4.8e-10	6.5
6	8^2	1.0e-6		2.0e-8		2.0e-8	
6	16^2	5.4e-8	4.3	2.4e-10	6.4	2.4e-10	6.4
6	32^2	3.2e-9	4.1	3.7e-12	6.0	3.7e-12	6.0
7	8^2	6.5e-7		2.1e-9		2.1e-9	
7	16^2	4.2e-8	3.9	5.5e-12	8.6	5.5e-12	8.6

Table 1: State L^2 error and order, sensor (R).

5. What the cutoff does *not* settle.

- *Under-resolution.* $N = 5$ on 4^2 (20 points per side, the least resolved run): $r_e > 10^{-3}$ in places, $\mu|_e$ up to $9 \cdot 10^{-5}$, error 3.6× Galerkin's. There the viscosity acts, as it should on a marginal wave.
- *Shocks.* $c = 10^{-3}$ is above this wave's floor on every mesh (at most $7 \cdot 10^{-4}$) and 5 decades below a shock's r_e, but it has not been run on the shock cases that use (R) (Brio–Wu, jet). With (L) (Orszag–Tang, rotor, blast, flux emergence) $r_e \approx \frac{3}{2}|\partial_t q|/\sigma = O(1)$ wherever the flow moves, so 10^{-3} would change little there. On the smooth MHD vortex ($N = 6$, 32^2, $t = 0.5$) the cutoff likewise gives Galerkin bit for bit; there the excess without it was only 2%.
- *What 10^{-3} means.* c is compared with r_e, the residual *divided by* σ_i, so it is a rate in the problem's time units: another normalization (ϵ_{rel}, element norms, `:dsgs_fast_floors`, which puts the fast speed in place of $\bar{c}$) or another time scale changes the threshold it stands for.

6. Where the paper's text and the code differ, for the quantities of this note.

- (11) and "the spatial part is read from the right-hand side the solver has just assembled" are the assembled rate of (L). The runs with (R) (smooth vortex, Brio–Wu, jet, this wave) use the element's own $M_e^{-1}\mathcal{R}_e$ and compute $\mu|_e$ once per step; the runs with (L) (Orszag–Tang, rotor, blast, flux emergence) evaluate (11) at every stage.
- (16) gives $\epsilon_{\text{rel}} = 10^{-3}$. The domain-norm decks run $\epsilon_{\text{rel}} = 1$ (`:dsgs_rel`); the flux-emergence decks normalize per element instead, floored at the element's own ρ, ρc, ρc^2, $\sqrt{\rho}\,c$.
- *Asymptotic behavior*: $\mu_{\text{res}} = O(h^{N+2}) + O(\Delta_e^2 \Delta t^2)$, and the second term is $O(h^4)$ at a fixed Courant number (§3), so without the cutoff DynSGS caps the order at 4.
- *Smooth vortex*: DynSGS on top of Galerkin up to $N = 4$, diverging above as the mesh is refined, earlier for higher N, is the signature of a time-difference floor of the sensor (§3), which the paper attributes to Gibbs modes.

Reproduce. `tools/alfven_wave_scan.sh` runs the deck as committed, (R) with $c = 10^{-3}$, for $N = 3 \ldots 7$ (`AW_JOBS=5` runs the five orders at once). With `AW_CUTOFF = 0.0` in `user_inputs.jl`, `AW_KEEP=1 AW_VISC=dsgs tools/alfven_wave_scan.sh` adds the curves without the cutoff (records `dsgs_nocut`); `JEXPRESSO_AW_SENSOR=legacy` selects (L). `dsgsCutoff_fig.py` redraws Fig. 1 from the case's error store.